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ORIGINAL RESEARCH

Multi-feature fusion attention network for single image super-resolution

Jiacheng Chen | Wanliang Wang | Fangsen Xing | Hangyao Tu

College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou, China

Correspondence
Wanliang Wang, College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou 310023, China.
Email: zjutwwl@zjut.edu.cn

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Abstract

Single Image Super-Resolution algorithms have made enormous progress in recent years. However, many previous Convolution Neural Network (CNN) based Super-Resolution algorithms only stack uniform convolution layers of fixed kernel size, and frequently ignore inherent multi-scale properties of the images, resulting in unsatisfactory reconstruction results. Here, a multi-feature fusion attention network (MFFAN) is proposed for capturing information at diverse scales. MFFAN is composed of multiple efficient sparse residual group (ESRG) modules. Several multi-scale feature fusion blocks (MSFFB) are constructed using a cascade manner in each ESRG module and it is capable of exploiting various cross scales information. Subsequently, a local-global spatial attention block (LGSAB) is inserted at the tail of the ESRG module for further improving the interaction of inter-pixel, which strengths essential features and suppresses irrelevant information. Additionally, owing to the fact that only feeding final output into the reconstruction layer has exacerbated the long-range dependency problems, an enhanced hierarchy feature fusion block (EHFFB) is designed to fuse low-level information and high-level semantic information. Experiment results indicate that the proposed MFFAN is competitive in comparison to several state-of-the-art algorithms.

1 | INTRODUCTION

Restoring degraded low-resolution image into a corresponding high-resolution image is known as Single-Image Super-Resolution (SISR). SISR [2] has become a research hotspot in academia and industry due to its wide-ranging applications, for instance, medical imaging [3], remote sensing imaging [4], and face recognition [5]. SISR is a typical ill-posed problem that low-resolution images lose the corresponding high-frequency features. A low-resolution image is degraded from several different high-resolution images, which poses a challenge for SISR tasks.

Recently, due to the rapid growth of deep learning, SISR algorithms [51] based on deep learning methods have substantially outperformed traditional methods, i.e., sparse code [6, 7] linear or nonlinear regression [8], and random forest [9]. Generally, existing deep-learning-based single image super-resolution algorithms are broadly characterized into PSNR metric-oriented and perceptual-oriented algorithms. Dong [10] firstly introduced convolution neural network (CNN) into super-resolution task and constructed a shallow network with only three convolution layers (SRCNN). Since then, CNN has dominated the SISR field. Although images recovered by these algorithms obtained high performance indicators, they are blurred and have less high-frequency information, which does not satisfy the human eye perception. To address this issue, Christian Ledig et al. [14] and Lin [1] proposed an SR algorithm using generative adversarial networks, which is capable of developing more visually pleasing images with powerful generative and fitting capabilities of GANs. Nevertheless, CNN-based algorithms outperformed this GAN-based methods in terms of quantitative performance. In addition, GAN-based methods often produce unrealistic images. Based on the fact that, many previous methods using a single size Convolutional kernel lack the ability to explore multi-scale features. And numerous identical patches reoccur inside the image itself, both within the same scale and across sizes. As a result, we propose a multi-scale feature fusion block to explore multi-scale features by employing convolutional operation with different kernel size, at the same time. In addition,

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to effectively fuse information at different scales, a channel attention mechanism is introduced. We discovered that the ease of recovery of images varies within an image. Consequently, we propose a novel local-global spatial attention block from both global and local perspectives to distorts resources toward patches that are not readily recoverable. As the network deepens, shallow and intermediate layer information is continuously processed by the convolutional operations, and rich low-level semantic information is constantly lost. Nevertheless, this low-level semantic feature is especially crucial for the reconstruction task. And previous methodologies simply impose all intermediate layer's information and output final extracted high-level semantic information through 1×1 convolution layer. The simple and brutal approach of multi-level information fusion ignores the inherent distinct information among different layers, which eventually impairs the expressiveness and performance of the final network. So, an enhanced hierarchy feature fusion net (EHFFN) is proposed to gradually perform features fusion among adjacent layers. Finally, we also propose an enhanced upscale network to further refine restored images in high-resolution space.

Our contributions are concluded as follows:

1. An efficient multi-scale feature fusion block (MSFFB) that captures a range of characteristics at multiple scales and significantly boosts the representation capability of the network is proposed.
2. We present a novel local-global spatial attention block (LGSAB) and effectively extract local and global spatial features, enabling the network to focus on high-frequency spatial information that is more relevant for recovering images.
3. A highly efficient multi-layer feature fusion approach is proposed named Enhanced Hierarchy Feature Fusion Block (EHFFB), EHFFB gradually integrates features among different layers effectively. In this scenario, we can greatly alleviate long dependency and gradient disappearance problems.
4. The newly proposed MFFAN achieves competitive performance in terms of both objective and subjective quality metrics on several common public datasets. The effectiveness of MFFAN has been confirmed by adequate experiments and ablation studies.

The remainder of this paper is organized as follows. Some Related works about SISR, Attention Mechanism, and Multi-Scale Extractor Block are briefly introduced in Section 2. Section 3 describes the proposed MFFAN in detail. Experiment results and ablation study are presented in section 4. Finally, the conclusion is drawn in section 5.

2 | RELATED WORKS

2.1 | CNN-based SISR methods

SISR has made unprecedented progress due to the rise of deep learning, especially the development of convolutional neural networks. Dong et al. [10] introduced the convolution neural network to the SISR task, using three CNN layers (SRCNN) to complete the mapping of low-resolution images to high-resolution images. Due to the powerful fitting and representation capabilities of CNN, SRCNN was substantially surpassing traditional SISR algorithms. Since then, various deep CNN architectures had been developed for solving SISR tasks. As deep CNNs obtained larger receptive fields and more contextual information was be captured, Kim et al. [11] constructed a deep convolutional network with 20 layers by introducing residual learning [15]. In addition, Kim et al. [16] and Tai et al. [17] used recurrent mechanisms and parameter sharing schemes to reduce the number of parameters so that they built deep networks without increasing computational overhead. The above methods first interpolated low-resolution image to corresponding high-resolution image and then put the interpolated image into network to produce final high-resolution result. Such a step would not only introduce additional artifacts and amplify the noise but also increase the cost of the calculation. To overcome the drawbacks above, FSRCNN [18] and ESPCN [19] input the LR images directly into the network and use a deconvolution [20] layer or sub-pixel [19] layer at the end of the network to amplify low-resolution images to corresponding resolution, which can reduce a substantial amount of computation. Deeper networks usually have better learning ability, so researchers usually construct deeper and wider networks to obtain more informative high-level features and thus strive for better performance. EDSR [12] built a 50+ layer deep network by removing BN [21] layers that are not applicable to low-level tasks. And, Zhang et al. [13] proposed a residual dense network (RDN) which used densely connected convolution layers to extract abundant local features and adaptively fused all hierarchical features in the LR space. Furthermore, Zhang et al. [22] proposed a model with more than 400 layers with an improved residual channel attention block. In order to combine the benefits of both residual and dense skip connections for enhanced network performance, Huang et al. [54] developed a dual-path attention network. To align several low-resolution burst images, Luo et al. [53] developed a feature enhanced pyramid cascading and deformable convolution module.

2.2 | Attention mechanism

Attention mechanisms are inspired by the functioning of human visual cortex cells to enhance salient information and suppress irrelevant information. It was initially proposed by Sutskever et al. [23] in machine translation via giving different weights to the input sequence. Hu et al. [24] introduce an attention mechanism to visual tasks by re-scale feature maps with a gating function (e.g. Sigmoid or Softmax). They propose a 'squeeze-and-excitation' module to learn interdependence between different channels. Furthermore, Zhang et al. [22] firstly introduce an attention mechanism to SISR task, and they use an attention block similar to SE-Net by considering inner-channel information, which calls first-order statistics. In contrast, Dai et al. [26] explore second-order information by

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calculating inter-channel corrections to learn discriminative feature expression and capture long-term spatial contextual information. Since these methods only consider interaction between channels, WOO et al. [25] propose an attention module for fusing channel and spatial information, named CBAM. Hu et al. [27] propose a channel-wise and spatial feature modulation to capture channel and spatial information simultaneously, making the network focus on high-frequency features. In addition, Zhang et al. [28] also propose a non-local attention mechanism that can capture long-term information. Nevertheless, the above approaches only consider channel attention or spatial attention, thus limiting the representational ability of networks. Niu et al. [29] propose a holistic attention network to model the holistic interdependencies among layers, channels, and positions. Furthermore, Yang et al. [30] propose a CASGCN block that is consists of several channel attention and spatial graph blocks to incorporate global spatial and channel correlations for deriving the features of each pixel. Mei et al. [52] propose a cross-scale non-local attention module to mine similarities between different scales in order to improve the reconstruction performance of the network. Zhang et al. [55] develop a context reasoning attention network (CRAN) which can adjust convolution kernel based on the global context information. It has been experimentally verified that this can greatly improve the generalization performance of the network.

2.3 | Multi-scale extraction block

Recently, single-image super-resolution networks based on deep CNNs have made great advancements, yet they often resort to deepening network depth or widening width of the network to improve final model performance, ignoring multi-scale relationships of various features. Firstly, GoogLeNet [31] puts several different convolutional kernels into same layer to extract information at different scales. Then, Fan et al. [32] designed a SISR network using a multi-scale feature extraction module as a backbone to assist to recover complex information. Multi-scale Residual Network [33] adaptively extracts features at different scales using 3×3 and 5×5 convolutional kernels, respectively. The upgraded version of MSRN, Multi-scale Dense Cross Network [34], introduces dense connection on the foundation of MSRB to improve feature reuse and information flow so it can extract rich features at different scales and levels, which is beneficial for high-frequency information restoration. Xiong et al. [35] propose an attention augmented multi-scale network (AAMN) to competently extract deep features from different scales which facilitates the reduction of network depth and lowers training difficulty. In addition, Lu et al. [36] propose a new multi-scale information polymerization network based on a multi-scale information polymerization block that effectively polymerizate the extracted features to obtain fine image features. To significantly reduce the number of parameters, a new feature extraction module, called asynchronous multi-scale block (AMB), is proposed by Jiahuan et al [37]. However, the above multi-scale methods tend to ignore the interaction between different convolution blocks at the same layer.

3 | MULTI-FEATURE FUSION ATTENTION NETWORK

In this section, the framework of MFFAN is introduced, shown in Figure 1. Then, we introduce four main components, multi shallow feature extract network (MSFEN), deep feature extract network (DFEN), enhanced hierarchy feature fusion network (EHFFN), and enhanced upscale network (EUN).

3.1 | Network framework

The framework of our proposed multi-feature fusion attention network is shown in Figure 1, which is composed of four components: (1) Multi shallow feature extract network (MSFEN), (2) deep feature extract network (DFEN), (3) enhanced hierarchy feature fusion network (EHFFN), and (4) enhanced upscale network (EUN). Input and output of our MFFAN are denoted by I_{LR} and I_{SR}, respectively. In previous literature [22, 26, 27], they extract shallow features from input images only by a single 3×3 convolution layer. Thus, they neglect different scales' information, presented in LR images. Therefore, we use three 3×3 dilated convolution [38] layers with dilation rates of 1, 3, and 5 for discovering abundancy multi-scale low-level features.

$$F_{shallow_i} = Conv_{3_i}(I_{LR}), i = 1, 2, 3 \quad (1)$$

$$F_{shallow} = Conv_1([F_{shallow_1}, F_{shallow_2}, F_{shallow_3}]) \quad (2)$$

where $Conv_{3_i}$ represents 3×3 convolution layers with different dilation rates to exploit complementary various multi-scale information. Then, the ultimate shallow features are obtained by fusing the diverse scale features based on a 1×1 convolution layer. In addition, the MSFEN can effectively integrate multi-scale features for superior contextual representations and generate rich structure cues simultaneously within different receptive fields. Next, $F_{shallow}$ is transferred into the DFEN to extract deep high-level features. The deep feature is denoted by Equation (3).

$$F_{deep} = H_{DFEN}(F_{shallow}) \quad (3)$$

where H_{DFEN} represents the function of the deep feature extractor net which consists of n ESRGs for feature efficiency exploration. To make full use of the extracted multi-level features, intermediate features are fed into EHFFN which consists of various AFFAB for feature adaptive selection and fusion. The Proposed EHFFN is represented as:

$$F_j = H_{AFFAB}([F_i, F_{i+1}]) \quad (4)$$

$$F_{j+1} = H_{AFFAB}([F_{i+2}, F_{i+3}]) \quad (5)$$

where F_i and F_{i+1} are the features generated by adjacent blocks. The concatenated features are fed into AFFAB for feature adaptive selection and fusion. We use four ESRGs to produce two enhanced features. Then, the above steps are repeated until only one enhanced feature is generated. Furthermore, the

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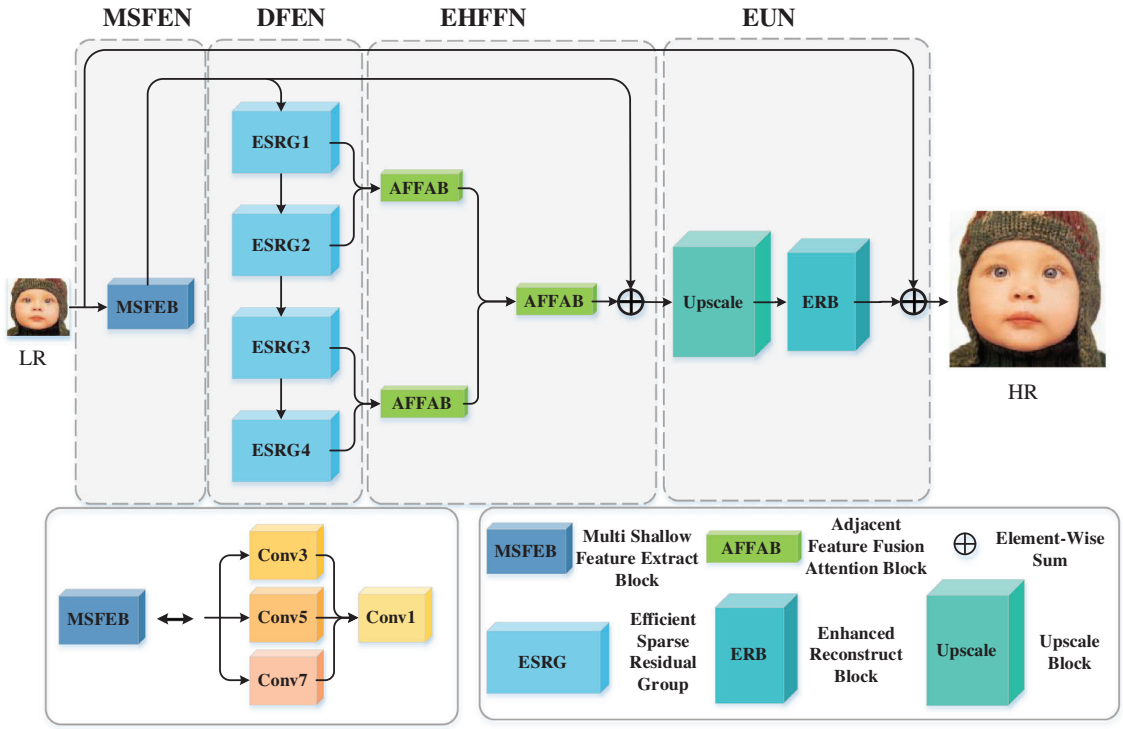


FIGURE 1 The architecture of our multi-feature fusion attention network (MFFAN)

enhanced deep features are fed into the EUN for reconstructing I_{SR} .

$$I_{SR} = H_{EUN}(F_{enhanced} + F_{shallow}) + H_{bilinear}(I_{LR}) \tag{6}$$

where H_{EUN} denotes up-sample module to scale up LR image to desired high-resolution through $F_{enhanced}$, $F_{shallow}$, and I_{LR}.$H_{bilinear}$ represents bilinear interpolation function. To further stabilize training progress of sophisticated deep network, a long skip connection (LSC) is introduced in our DFEN, which enhances convergence speed and improves the performance of the MFFAN network. In addition, LSC can encourage the flow of low-level information to deep layers, so that only high-level semantic information is focused on. To obtain the more realistic upscaled feature, we use a sub-pixel layer that demonstrates high performance with low computational complexity to reconstruct LR images. Finally, the proposed MFFAN is formulated as:

$$I_{SR} = H_{MFFAN}(I_{LR}) \tag{7}$$

To optimize SISR network, various loss functions have been investigated, such as $L1$ loss [12], $L2$ loss [17], perceptual loss [14], and adversarial loss [14]. As stated by Bee Lim et al. [12], $L2$ loss can maximize peak signal-to-noise (PSNR) metrics, but it prone to produce blurry images. Therefore, $L1$ loss is chosen as our optimization function for training the MFFAN. Then, the MFFAN is optimized by minimizing the pixel-wise dissimilarity between the estimated super-resolved image SR and corresponding ground truth HR. The optimization function is formulated as:

$$L(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\left\|H_{MFFAN}(I_{LR}^{i}) - I_{HR}^{i}\right\|_1 \tag{8}$$

where Θ denotes the trainable parameters of our MFFAN network. And this deep MFFAN is trained by using a training set $\{I_{LR}^{i}, I_{HR}^{i}\}_{i-1}^{N}$, which contains N LR images and their HR counterparts. Next, each key element of the MFFAN is described.

3.2 | Deep feature extractor network

In this section, the proposed Deep Feature Extractor Network (DFEN) is introduced, which consists of several ESRG modules, as shown in Figure 1. Next, each key component of the ESRG module is described in detail. MSFFB is in section 3.2.2 and LGSAB is in Section 3.2.3.

Inspired by Zhang et al [22], we build our DFEN using the Residual in Residual (RIR) structure, which is very effective in building the very deep network to alleviate the problem of exploding or vanishing gradients. The gth ESRG is formulated as:

$$F_g = H_g(F_{g-1}) = H_g(H_{g-1} \ldots H_1(F_{shallow})) \tag{9}$$

where H_g indicates the function of gth ESRG. F_{g-1} and F_g are input and output of gth ESRG respectively.

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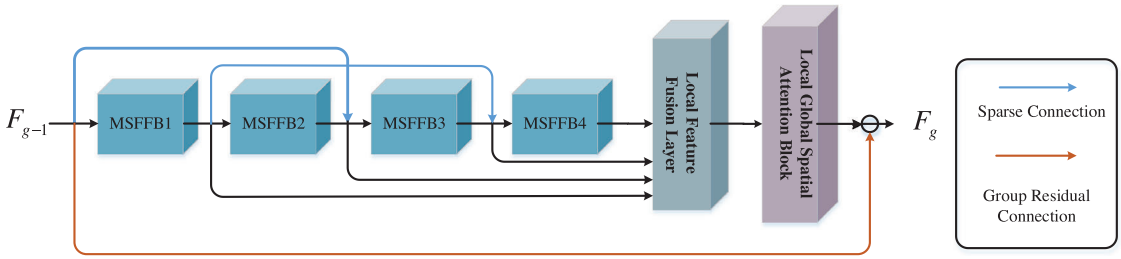


FIGURE 2 Efficient sparse residual group (ESRG)

3.2.1 | Efficient sparse residual group

As LR images and HR images are highly related, they contain a significant amount of comparable information, so MFFAN only needs to retrieve lost high-level information. To allow more abundancy low-level information stream to deep layers, we cascade 4 multi-scale feature fusion blocks (MSFFB), a local feature fusion layer (LFFL), and a local-global spatial attention block (LGSAB) in each ESRG, as illustrated in Figure 2.

$$F_{g,b} = \begin{cases} H_{g,b}(F_{g,b-1}) & b = 1,2 \\ H_{g,b}(Conv_1([F_{g,b-1}, F_{g,b-3}])) & b = 3,4 \end{cases} \tag{10}$$

where $H_{g,b}$ is gth MSFFB of ESRG. $Conv_1$ denotes a 1×1 convolution layer to fuse distinct hierarchical features. This sparse connection pattern is unlike the dense connection used by RDN [13] which requires huge computing resources and storage space. Furthermore, it can reuse features and promotes the efficiency of our network without sacrificing computational capacity.

$$F_{g,fuse} = Conv_1([F_{g,1}, F_{g,2}, F_{g,3}, F_{g,4}]) \tag{11}$$

$$F_{g,sa} = H_{LGSA}(F_{g,fuse}) \tag{12}$$

Local feature fusion is implemented by a 1×1 convolution layer, then merged features are delivered into LGSAB module to further improve MFFAN's representation capacity.

3.2.2 | Multi-scale feature fusion block

As previously stated, existing many super-resolution algorithms based on CNNs lack the capacity to explicitly investigate multi-scale features efficiently, resulting in SR network performance degradation. Inspires by Inception network [31], MSRN [33] a unique and efficient multi-scale feature extraction module is proposed, MSFFB, as shown in Figure 3. WDSR [39] states that widening the width of the network before activation function allows more useful information to deep layers, so this paper uses a 1×1convolution layer to expand channels of features.

$$F_{\exp and} = Conv_1(F_{in}) \tag{13}$$

$$F_i = Conv_{1_i}(F_{\exp and}) i = 1,2,3 \tag{14}$$

$$F_{3_1} = Conv_{3_1}(F_1) \tag{15}$$

$$F_{3_2} = Conv_{3_2}([F_2, F_{3_1}]) \tag{16}$$

$$F_{3_3} = Conv_{3_3}([F_3, F_{3_2}]) \tag{17}$$

$$F_{out} = CA([F_{3_1}, F_{3_2}, F_{3_3}]) + F_{in} \tag{18}$$

MFFAN uses 1×1 convolution to generate three inputs for three different 3×3 convolution layers. $Conv_{3_i}$ represents 3×3 convolution layer with expansion rate i to extract different scale features. And [] represents concat operation. It is noted that the proposed MSFFB adopts several parallel but connected paths to aggregate multi-scale features progressively. In addition, the features extracted by small convolution kernels are used to assist large convolution kernels to extract features. The retrieved multi-scale characteristics are then transmitted to a channel attention module to boost MFFAN's discriminative capacity even further. Rather than existing methods that employ a simple concatenation or averaging of features coming from multi-scale branches, a channel attention (CA) method is used to dynamically select the most beneficial collection of features. CA mechanism provides a way to adaptively reweight feature maps by exploiting the dependencies among different channels in a global way. In this way, we can achieve mutual gains between various multi-scale features. Let $F^{C \times H \times W}$ denotes input of CA module. A channel-wise descriptor $Z \in j^{1 \times 1 \times C}$ is obtained by using a global average pooling operation. Then the cth element of Z is determined by:

$$Z_C = \sum_{i=1}^{H} \sum_{j=1}^{W} F^C(i,j) \tag{19}$$

where $F^C(i,j)$ is the value at the position (i,j) of cth channel of F^C. Then a gating mechanism is used to aggregate information among channels.

$$att_map = \sigma(W_U \delta(W_D(Z))) \tag{20}$$

where W_D denotes the weight of the first FC layer, which acts as channel dimension reduction from C toC/r. Conversely, W_U increases the channel dimension back to C.r means reduction ratio. δ and σ represent ReLU function and Sigmoid function respectively. More specifically, MSFFB uses the attention map

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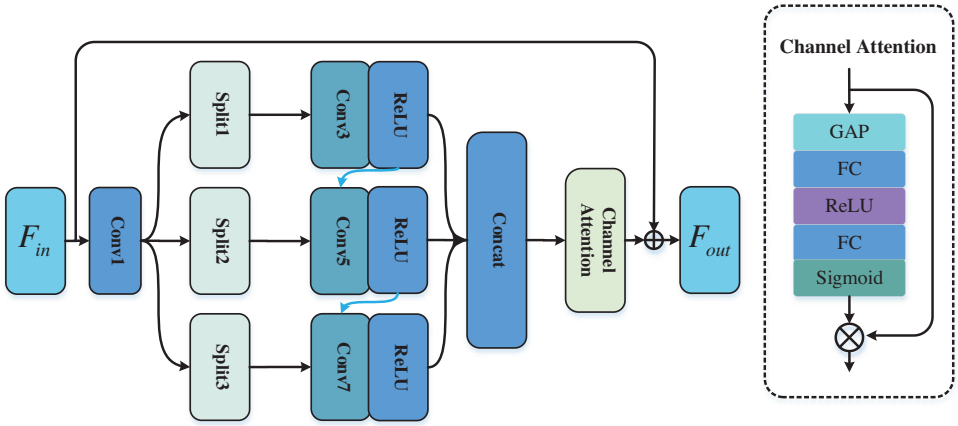


FIGURE 3 Multi-scale feature fusion block

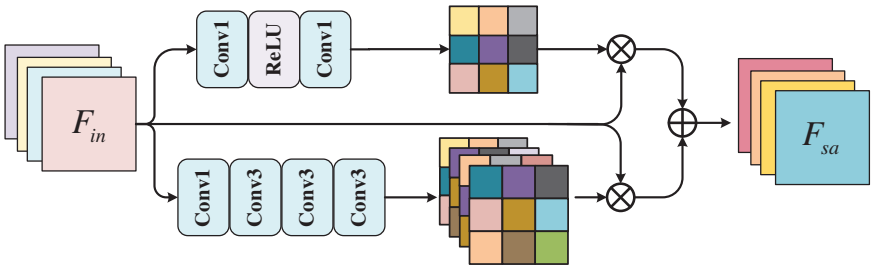


FIGURE 4 Local-global spatial attention block

to rescale input features.

$$F_{out} = F_{in} \otimes att_map + F_{in} \tag{21}$$

here $\otimes$ represents the element-wise multiplication and residual learning is introduced to stabilize the training of our MFFAN.

3.2.3 | Local-global spatial attention block

There are several types of information in nature photographs, ranging from more intricate texture and edge information to flat and basic backdrop information. As a result, we must improve the network's capacity to discern areas of differing complexity. Channel-wise and spatial feature modulation network [27] proposes a spatial attention module that focuses only on global information and ignores the spatial difference between channels, which affects the final performance of the network. Therefore, a novel local-global spatial attention module (LGSAB) (as shown in Figure 4) is presented that is capable of adapting to the complexity of different regions. LGSAB aims to recalibrate complementary contextual features in a local–global manner using the relationships between different separated areas. Furthermore, the LGSAB mechanism operates towards the conclusion of the ESRG to drive features to be more concentrated on the regions of interest. LGSAB is composed of two branches, the top one is responsible for detecting global spatial features and the bottom one is in charge of detecting spatial features at each channel.

$$F_{sa_global} = \sigma\left(Conv_1^{c/r,1}\left(\delta\left(Conv_1^{c,c/r}(F_{in})\right)\right)\right) \times F_{in} \tag{22}$$

$$F_{sa_local} = \sigma\left(Conv_1^{c/r,c}\left(Conv_{3,d=2}^{c/r,c/r}\left(Conv_{3,d=2}^{c/r,c/r}\left(Conv_1^{c,c/r}(F_{in})\right)\right)\right)\right) \times F_{in} \tag{23}$$

$$F_{sa} = F_{sa_global} + F_{sa_local} \tag{24}$$

where σ represents a sigmoid function, $Conv_{k,d}^{c_in,c_out}$ indicates $k \times k$ kernel size, and c_in, c_out represent input channel and output channel respectively. LGSAB uses a 1×1 convolutional layer to reduce channel dimensions so that the whole local SA branch can be greatly lightweight. In addition, a large receptive field is required for attention block to work well for the task of image SR. Thus, we use dilated convolution [38] with an expansion rate of d to improve the perception field without losing information.

3.3 | Enhanced hierarchy feature fusion network

In this section, we go through the EHFFN, which is made up of multiple AFFABs. As described in the previous section, the fusion of hierarchical information only by one 1×1 convolution layer is prone to damage the expressive power of the network, as hierarchical features often contain different semantic information. Shallow features have more texture and edge information than higher-level features, which are more abstract. Therefore, an AFFAB (as shown in Figure 5) is developed

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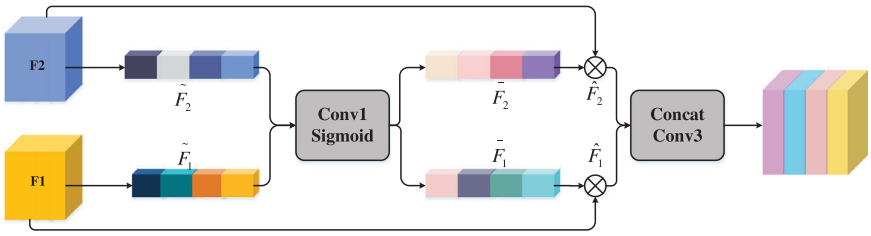


FIGURE 5 Adjacent feature fusion attention block

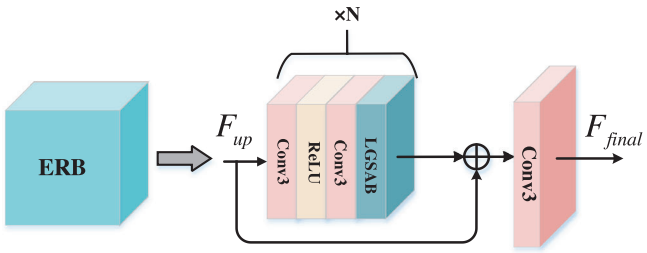


FIGURE 6 Enhanced reconstruction block

to gradually integrate hierarchical features. Moreover, AFFAB employs a selective network that dynamically combines variable-level characteristics dynamically while correctly preserving the important features at each level. Let F_1 and F_2 denote two adjacent features with the size of $C \times H \times W$ respectively. Then, to generate channel-wise statistics $\tilde{F}_1$ and $\tilde{F}_2$, the global pooling is operated with $H \times W$ dimension, as done in [23].

$$\tilde{F}_n = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_n^C(i,j) n = 1,2 \tag{25}$$

$$\bar{F}_n = \delta(Conv_1(\tilde{F}_n)) n = 1,2 \tag{26}$$

$$\hat{F}_n = \bar{F}_n \otimes F_n n = 1,2 \tag{27}$$

$$F_{enhanced} = Conv_1([\hat{F}_1, \hat{F}_2]) \tag{28}$$

where $F_n^C(i,j)$ is the value at the position (i,j) of cth feature F_n. Then a gating function is introduced to learn the channel-wise inter-dependencies. δ denotes Sigmoid function. $Conv_1$ is the 1×1convolution layer and [] represents concat operation. $\otimes$ represents element-wise multiplication. In this way, we can automatically select and integrate the most useful information.

3.4 | Enhanced reconstruction network

In previous methods [22, 26, 27], the most researchers only used one convolution layer to expand channels of deep feature, and then used a sub-pixel layer [19] to expand images to the appropriate scale. We state that this method is too simple that it seldom recovers high-frequency details of images well and ultimately hinders the network for recovering images. Furthermore, all features are extracted in LR feature space and feature relationships in HR space are severely missing, resulting in blurred recovery images. In order to fit the HR space, a novel enhanced reconstruction block is proposed to enhance feature relevance in HR space, as shown in Figure 6.

$$F_{final} = H_{lgsab}(Conv_3(\delta(Conv_3(\ldots H_{lgsab}(Conv_3(\delta(Conv_3(F_{up}))))))))) \tag{29}$$

where F_{up} and F_{final} are input and output of ERB respectively. Specifically, the ERB module consists of N convolutional modules, each consisting of two convolutions, an activation function and a LGSAB. H_{lgsab} is the local-global spatial attention block. $Conv_3$ and $\delta$are the convolution layer and ReLU activation function respectively. Since the HR information and LR information are captured at the same time, our network achieves better performance.

4 | EXPERIMENTS AND ANALYSIS

4.1 | Datasets and implementation

For a fair comparison, DIV2K [40] dataset is used which has 800 training images, 100 validation images to train our deep network. In the testing stage, this paper use five commonly employed test sets, including Set5 [41], Set14 [42], BSD100 [43], Urban100 [44], and Manga109 [45], to measure MFFAN and compare it with other state-of-the-art methods. To obtain better generalization performance, we use data augmentation, which includes random rotation by 90°, random horizontal flipping and vertical flipping. Peak signal-to-noise (PSNR) and structural similarity (SSIM) are our performance metrics. A higher metric indicates a superior quality of recovered images. For training, we randomly crop 16,000 48 × 48 LR patches as the training set, and batch size is set to 16. The learning rate is initialized to 1×10^{-4}, which is halved at every 200 epochs. This paper uses the Adam [46] optimizer by setting $\beta_1 = 0.9$, $\beta_2 = 0.9999$ and $\varepsilon = 10^{-8}$. In our MFFAN model, all convolution layers contain 64 filters except the 1×1 layers. Finally, eight ESRGs are used to form our DFEN, and each ESRG includes four MSFFB modules.

4.2 | Comparisons with the state-of-the-art methods

4.2.1 | Quantitative analysis

To demonstrate the effectiveness and superiority of this paper proposed algorithm (MFFAN), several state-of-the-art SISR methods, including Bicubic, SRCNN [10], FSRCNN [18], VDSR [11], DRRN [17], MemNet [47], IDN [48], EDSR [12], RDN [13], and CSNLN [52] are compared in terms of

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TABLE 1 The PSNR/SSIM results on tests of scales × 2, × 3, and × 4. The best and second results are highlighted and underlined

		Set5		Set14		B100		Urban100		Manga109	
Method	**Scale**	**PSNR**	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**	**PSNR**	**SSIM**
Bicubic	×2	33.68	0.9304	30.24	0.8691	29.56	0.8435	26.88	0.8405	30.81	0.9348
SRCNN [10]	×2	36.66	0.9542	32.45	0.9067	31.36	0.8879	29.51	0.8946	35.70	0.9677
FSRCNN [18]	×2	36.98	0.9556	32.62	0.9087	31.50	0.8904	29.85	0.9009	36.56	0.9703
VDSR [11]	×2	37.53	0.9587	33.05	0.9127	31.90	0.8960	30.77	0.9141	37.41	0.9747
DRRN [17]	×2	37.74	0.9591	33.23	0.9136	32.05	0.8973	31.23	0.9188	37.88	0.9750
MemNet [47]	×2	37.78	0.9597	33.28	0.9142	32.08	0.8978	31.31	0.9195	38.02	0.9755
IDN [48]	×2	37.83	0.9600	33.30	0.9148	32.08	0.8985	31.27	0.9196	38.02	0.9749
EDSR [12]	×2	38.11	0.9601	33.92	0.9195	32.32	0.9013	32.93	0.9351	39.19	0.9782
RDN [13]	×2	38.24	0.9614	34.01	0.9212	32.34	0.9017	32.89	0.9353	39.18	0.9780
CSNLN []	×2	38.28	0.9616	34.12	0.9223	32.40	0.9024	**33.25**	**0.9386**	39.37	0.9785
MFFAN (ours)	×2	**38.29**	**0.9617**	**34.24**	**0.9248**	**32.41**	**0.9025**	33.09	0.9361	**39.45**	**0.9787**
Bicubic	×3	30.40	0.8686	27.54	0.7741	27.21	0.7389	24.46	0.7349	26.95	0.8565
SRCNN [10]	×3	32.75	0.9090	29.29	0.8215	28.41	0.7863	26.24	0.7991	30.56	0.9125
FSRCNN [18]	×3	33.16	0.9140	29.42	0.8242	28.52	0.7893	26.41	0.8064	31.12	0.9196
VDSR [11]	×3	33.66	0.9213	29.78	0.8318	28.83	0.7976	27.14	0.8279	32.13	0.9348
DRRN [17]	×3	34.03	0.9244	29.96	0.8349	28.95	0.8004	27.53	0.8378	32.74	0.9344
MemNet [47]	×3	34.09	0.9248	30.00	0.8350	28.96	0.8001	27.56	0.8376	32.79	0.9391
IDN [48]	×3	34.11	0.9253	29.99	0.8354	28.95	0.8013	27.42	0.8359	32.69	0.9378
EDSR [12]	×3	34.65	0.9282	30.52	0.8462	29.25	0.8093	28.80	0.8653	34.20	0.9486
RDN [13]	×3	34.71	0.9296	30.57	0.8468	29.26	0.8093	28.80	0.8653	34.13	0.9484
CSNLN	×3	34.74	0.9300	30.66	0.8482	**29.33**	**0.8105**	**29.13**	**0.8712**	34.45	0.9502
MFFAN (ours)	×3	**34.75**	**0.9397**	**30.93**	**0.8555**	**29.33**	0.8098	28.99	0.8678	**34.47**	**0.9503**
Bicubic	×4	28.43	0.8109	26.00	0.7023	25.96	0.6678	23.14	0.6574	24.89	0.7875
SRCNN [10]	×4	30.48	0.8628	27.50	0.7513	26.90	0.7103	24.52	0.7226	27.63	0.8553
FSRCNN [18]	×4	30.70	0.8657	27.59	0.7535	26.96	0.7128	24.60	0.7258	27.85	0.8557
VDSR [11]	×4	31.35	0.8838	28.02	0.7678	27.29	0.7252	25.18	0.7252	28.87	0.8865
DRRN [17]	×4	31.68	0.8888	28.21	0.7720	27.38	0.7284	25.44	0.7638	29.45	0.8946
MemNet [47]	×4	31.74	0.8893	28.26	0.7723	27.40	0.7281	25.50	0.7630	29.64	0.8971
IDN [48]	×4	31.82	0.8903	28.25	0.7730	27.41	0.7297	25.41	0.7632	29.41	0.8936
EDSR [12]	×4	32.46	0.8968	28.80	0.7876	27.71	0.7420	26.64	0.8033	31.03	0.9158
RDN [13]	×4	32.47	0.8990	28.81	0.7871	27.72	0.7419	26.61	0.8028	31.00	0.9151
CSNLN []	×4	32.68	0.9004	28.95	0.7888	27.80	0.7439	**27.22**	**0.8168**	**31.43**	**0.9201**
MFFAN (ours)	×4	**32.69**	**0.9008**	**29.18**	**0.8001**	**27.81**	**0.7440**	26.97	0.8093	31.34	0.9189

quantitative and visual quality. As in [12, 13, 22] SR images are converted to YCbCr space and then calculate PSNR and SSIM [49] on the luminance channel. For model evaluation, this paper applies the Bicubic interpolation operation to obtain LR images from original HR images. To make a fair comparison, we crop off s (scaling factor) pixels from the edges of the SR image. In addition, quantitative evaluations are made in five datasets for three scales (×2, ×3, and ×4). The experiment results are shown in Table 1. Specifically, on challenging dataset Manga109, this paper proposed MFFAN advances the state-of-the-art methods with improvement margins of 0.26, 0.33, and 0.26 dB on scale factors ×2, ×3, and ×4 respectively. To a large extent, this can be attributed to the fact that the Manga109 dataset contains sufficient texture features and the MSFFB module can sufficiently extract multi-scale texture information for recovering detailed information. Furthermore, we found that our network can achieve more outstanding performance on SSIM metrics. This is because the MFFAN is able to fully fuse low-level texture information with high-level semantic information to recover rich structural and granular information. Compared with MemNet [47] and RDN [13], both of which introduce extremely dense skip connections, the MFFAN network achieves the highest performance but with sparse skip connection which can significantly reduce the number of calculations. This proves that

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TABLE 2 The NIQE results on tests of scales ×2. The best and second results are highlighted and underlined

Dataset	EDSR NIQE↓	RNAN NIQE↓	RDN NIQE↓	MFFAN (ours) NIQE↓
Set5	<u>5.667</u>	5.875	5.801	**5.369**
Set14	5.122	5.149	**5.069**	<u>5.089</u>
Urban100	4.656	4.681	<u>4.626</u>	**4.599**
Manga109	4.683	4.600	<u>4.556</u>	**4.538**

TABLE 3 Comparison of models trained with different basic blocks

Basic block	Set5	Set14	B100	Urban100	Manga109
Residual block	37.962	33.531	32.143	31.975	38.31
Dense block	38.039	33.57	32.184	32.118	38.567
Sparse block	38.043	33.635	32.187	32.086	38.691

the ESRG block with sparse skip connections surpasses the memory block in MemNet [47] and the residual dense block in RDN [13]. Furthermore, compared to the current leading method CSNLN, our network achieves competitive results with less computational resources (Flops). In order to demonstrate that our method is also applicable in real-world scenarios, we further use the Natural Image Quality Evaluator (NIQE) to evaluate the performance of MFFAN. The lower the image quality is, the higher NIQE is. As seen in Table 2, MFFAN outperforms other approaches by a significant margin. Specifically, MFFAN is 0.432 lower than the current SOTA method RDN on the Set5 dataset. This well illustrates that our method can be applied not only to synthetic datasets, but also to real datasets.

4.2.2 | Qualitative analysis

In Figures 7–9, the visual comparison of multiple methods on the Urban100 dataset for scale factor 2 is shown. As shown, although little information has been lost in the ×2 degradation, the previous methods cannot recover lost high-frequency information well, resulting in blurring and distortion of the roofline. In contrast, this paper proposed MFFAN can recover high-frequency information (e.g. lines and line orientation) that is highly consistent with HR, and the whole image presents sharper texture details. In Figure 7, MFFAN is able to recover roof lines well, however, RDN and RCAN (the SOTA methods) cannot recover the direction of lines correctly and produce more blurred areas. In Figure 8, MFFAN is able to correctly recover edge information between glass windows, nevertheless, RNAN and RDN produce very severe blurring effects. Furthermore, RCAN cannot produce rails in the same direction as HR, but MFFAN can produce rails that are consistent with HR in Figure 9. This is mainly because the MSFFAB module we used is able to extract adequate information at different scales in texture-rich regions. Furthermore, Figures 10 and 11 show us the more challenging results with magnification factors of 3 and 4, respectively. In Figure 10, other models bring obvious artifacts and make restored images more ambiguous. In particular, the current mainstream models RDN and RCAN cannot correctly produce contours of the letter T. Nevertheless, the predictions of MFFAN are closer to the HR. In Figure 11, MFFAN is also able to produce the most regular window glass. On one hand, this is because the proposed MSFFB may use a wealth of information to determine which parts should be reserved or ignored, as a result, additional scale aspects will not conflict with it in an unanticipated way. On the other hand, RCAN and RDN are not good at processing more subtle structures, on the contrary, MFFAN can fully extract detailed information at different scales and levels. As a result, the proposed MFFAN is capable of distinguishing error-prone shapes from complicated structures and inferring precise and sharper features that better suit the HR.

4.3 | Study of ESRG block and the numbers of ESRGs

To confirm the superiority of Efficient Sparse Residual Group (ESRG) on the SISR reconstruction task (as shown in Table 3), two other general base modules are compared (ResBlock [15]

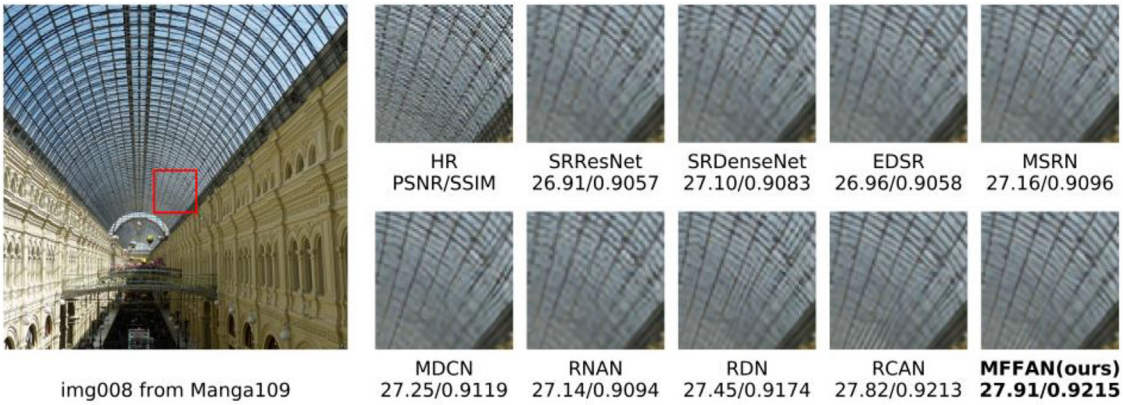


FIGURE 7 Visual comparison for ×2

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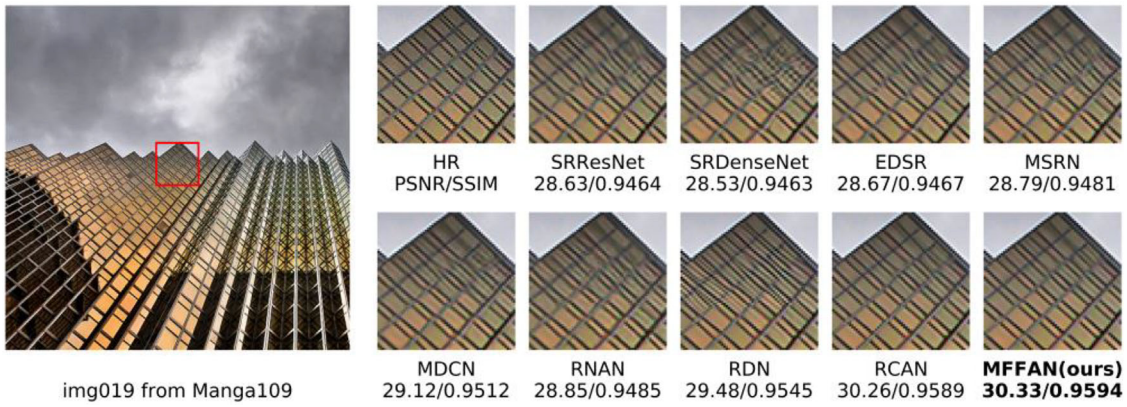


FIGURE 8 Visual comparison for ×2

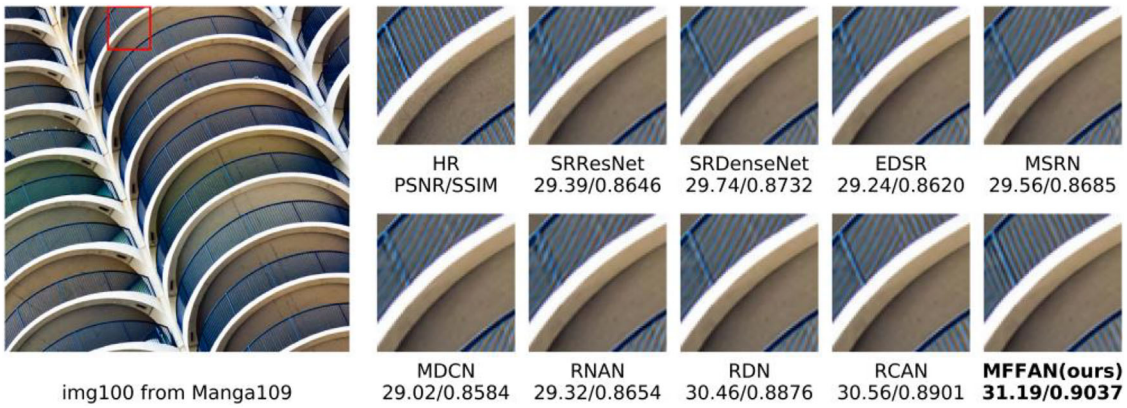


FIGURE 9 Visual comparison for ×2

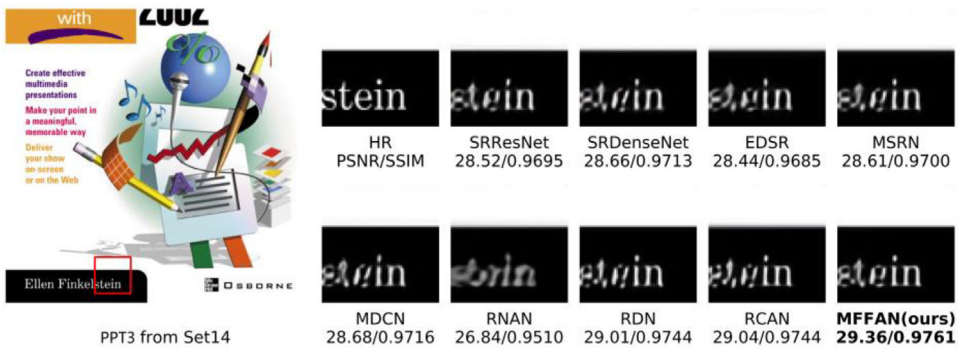


FIGURE 10 Visual comparison for ×3

and DenseBlock [50]). ResBlock [15] simply sums the input and output to get the final result, which often lose a significant amount of intermediate details, ultimately affecting the performance of the network. In contrast, DenseBlock [50] feeds intermediate information from all previous layers to later layers, which not only causes a great deal of information redundancy but also dramatically increases the computational load of the network. To overcome the aforementioned issues, an efficient ESRG is developed that combines the advantage of ResBlock and DenseBlock simultaneously. It should be emphasized that ESRG only transmits current information to more distant layers in an effective way. Table 3 shows a performance comparison between MFFAN with ResBlock, DenseBlock, or SparseBlock on benchmark datasets with the scale of 2. It can be found that SparseBlock achieves the best performance on almost all test sets. On the Manga109 dataset, an improvement of 0.283 dB is feasible, demonstrating the effectiveness of the proposed ESRG.

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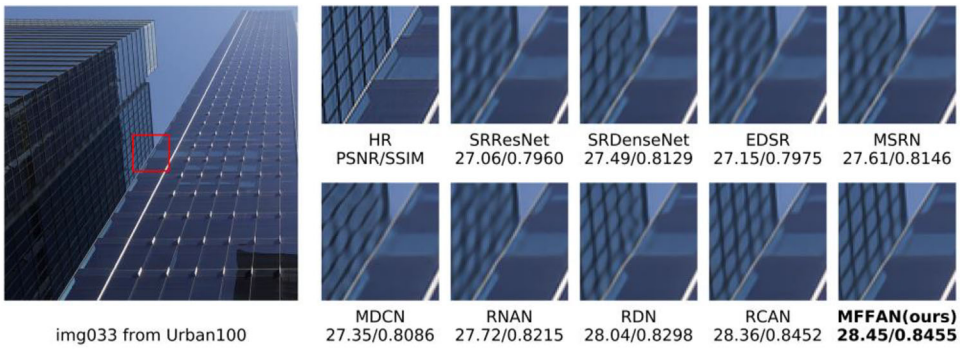
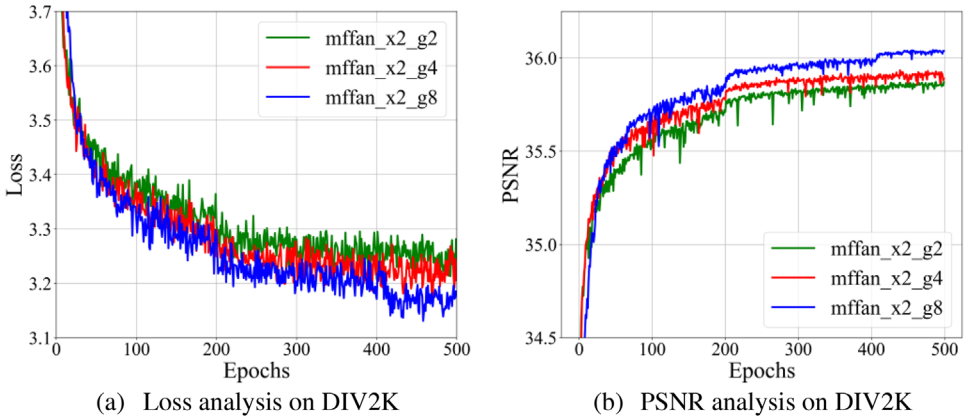


FIGURE 11 Visual comparison for ×4



(a) Loss analysis on DIV2K (b) PSNR analysis on DIV2K

FIGURE 12 performance analysis on DIV2K validation sets with scaling factor ×2. (a) Loss analysis on DIV2K, (b) PSNR analysis on DIV2K

In this subsection, the impact of the number of ESRG modules (g = 2, 4, 8) on the network performance, is explored since the depth of the network has a pivotal effect on model reconstruction properties. Figure 12 shows the performance analysis of this paper proposed MFFAN accompanied by the different number of ESRG modules on the DIV2K validation dataset with scaling factor 2, including loss function and reconstruction performance. As we suspected, the reconstruction performance of MFFAN improves as g becomes larger, since the complexity of the network and the feature extraction ability are enhanced. As a result, MFFAN with 8 ESRG modules can converge faster and achieve smaller loss. It is worth noting that when we train MFFAN with 16 ESRG modules, the network rapidly collapses and does not converge well, which is attributed to gradient explosion. So, we finally set g to 8 in all experiments.

4.4 | Study of LGSA block

An LGSA block is developed to eliminate redundant worthless characteristics and extract most essential features. In Table 4, Base method represents the model without any spatial attention mechanism, and GSA method represents the model with global spatial attention, which only has the upper branch in LGSA. Obviously, with the aid of spatial attention, the model performance can be further improved. The GSA mechanism boosts model average performance about 0.041 and 0.051 dB compared with the Base model for ×2 and ×4 scale factors respectively. This can be attributed to the ability of GSA to guide MFFAN with global spatial information to extract more useful features. Furthermore, using local spatial attention and global spatial attention simultaneously can contribute most (0.064 and 0.1126 dB) to the improvement of PSNR values and make the model training more stable. Therefore, we can conclude that LSA and GSA are complementary to each other.

TABLE 4 Comparisons on different spatial attention blocks

Method	Scale	Set5	Set14	B100	Urban100	Manga109
Base	×2	37.621	33.188	31.917	31.074	37.622
	×4	31.635	28.210	27.298	25.344	29.405
GSA	×2	37.657	**33.208**	31.938	31.130	37.694
	×4	31.667	28.271	27.340	25.391	29.478
LGSA	×2	**37.704**	33.206	**31.955**	**31.144**	**37.733**
	×4	**31.736**	**28.326**	**27.374**	**25.455**	**29.564**

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TABLE 5 Ablation investigation of different components

Components	Different combinations of components								
MSFF Block	✗	✓	✗	✗	✓	✓	✗	✓	✓
LGSA Block	✗	✗	✓	✗	✓	✗	✓	✓	✓
EHFF Block	✗	✗	✗	✓	✗	✓	✓	✓	✓
ERB Block	✗	✗	✗	✗	✗	✗	✗	✗	✓
PSNR (dB)	33.543	33.692	33.566	33.627	33.868	33.854	33.703	33.897	**33.940**

4.5 | Ablation study

In this subsection, the contributions of different components in our model are studied via ablation experiments. The module without MSFFB, LGSA EHFF, and ERB as our baseline model. All comparison models are trained for 500 epochs on the DIV2K dataset with the same experimental conditions. From Table 5, we can conduct that using MSFFB can improve 0.149 dB compared with the baseline model. The multi-scale feature fusion module can automatically explore and fuse features of various scales which play important roles in the reconstruction of degraded images. Compared with the baseline model, the LGSA mechanism achieves an improvement of 0.23 dB. The LGSA model allocates more resources to the more critical region and suppresses irrelevant information from both local and global viewpoints, allowing the network to pay more attention to high-frequency information. Furthermore, the proposed ERB achieves a 0.043 dB improvement and refines the reconstruction images from HR space, which complements the features extracted from LR space, thus improving the restoration performance. Overall, the performance of the entire network is significantly better than the network without our proposed components, revealing that the validity of our proposed module.

4.6 | Model size and execution time analysis

In this section, we study the trade-off of PSNR versus parameters, PSNR versus Flops and PSNR vs. execution time. The performance and model size of other classic deep CNN-based SISR approaches are shown in Figure 13. Obviously, MFFAN achieves competitive results with fewer parameters. VDSR, DRRN and MemNet, among these approaches, contain much less parameters at the expense of performance. Even though CSNLN has fewer parameters than MFFAN, its performance is lower by 0.12 dB. On the other hand, MFFAN contains fewer parameters than RDN, SAN and RCAN but achieve greater performance, implying that our MFFAN can achieve a good balance of performance and model complexity. Next, we study the trade-off between execution time and PSNR performance of different SOAT approaches. SAN and CSNLN have comparable result to our method, as demonstrated in Figure 14 but their execute time is three times longer than ours. Furthermore, with same inference time, MFFAN achieves better performance than RCAN, RDN and, EDSR. The performance of various

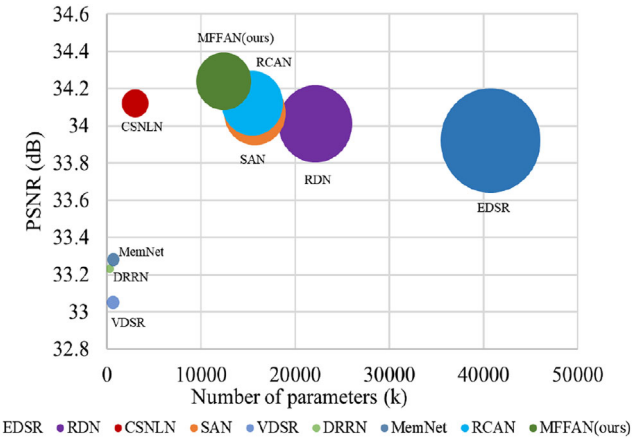


FIGURE 13 Model size and performance comparison on Set14 for ×2 SR

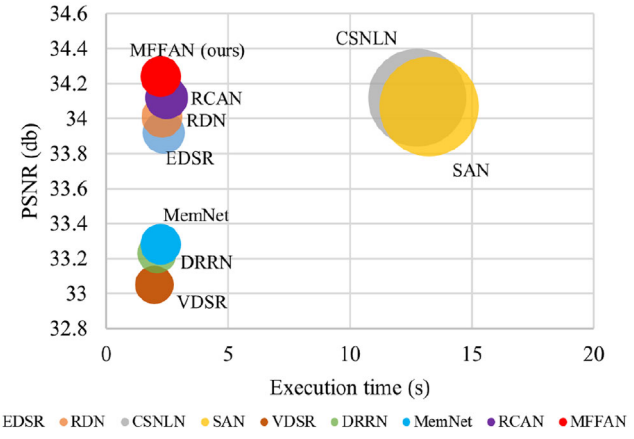


FIGURE 14 Trade-off between performance and execution time on Set14 for ×2 SR

SOAT algorithms' trade-off between Flops and PSNR is then investigated. As shown in Figure 15 CSNLN achieves results similar to our approach, although their Flops is over three times larger. In summary, compared with other approaches, MFFAN can greatly reduce parameters of the model and execute time while guarantying performance.

5 | CONCLUSION

Here, a multi-feature fusion attention network (MFFAN) is proposed for highly accurate single image super-resolution, which

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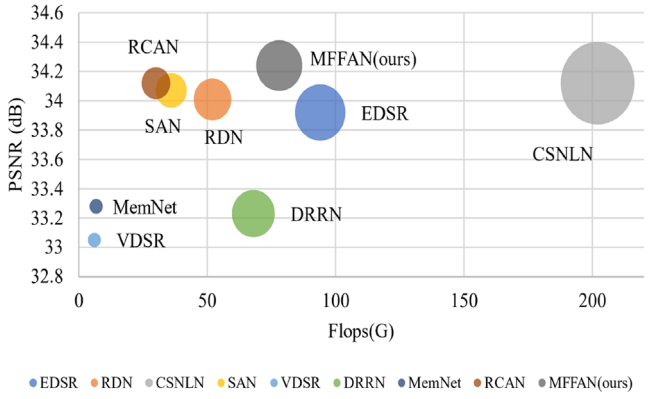


FIGURE 15 Model flops and performance comparison on Set14 for ×2 SR

utilizes efficient sparse residual group (ESRG) with local-global spatial attention block (LGSAB) to enhance feature extraction capability. Specifically, the ESRG consists of four multi-scale feature fusion blocks (MSFFBs) with sparse residual connections. In order to adequately capture multi-scale features, the MSFFB contains three dilated convolution layers with different dilated rates. In addition, an enhanced hierarchy feature fusion net is proposed to further boost the reconstruction power and discriminatory ability of the network, which can gradually integrate useful adjacent information. The experimental results of the single image super-resolution task with the BI degradation model illustrate the effectiveness of the proposed MFFAN network in terms of quantitative and visual results. In the future, we will adapt the proposed multi-scale feature fusion attention network to the real SISR tasks and exploit more powerful and highly efficient methods to conduct precise SISR reconstruction.

AUTHOR CONTRIBUTIONS

Jiacheng Chen: Investigation; Methodology; Software; Validation; Writing - original draft. Wanliang Wang: Funding acquisition; Project administration. Fangsen Xing: Software; Validation; Visualization. Hangyao Tu: Writing - review editing. All authors have read and agreed to the published version of the manuscript.

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CONFLICT OF INTEREST

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Jiacheng Chen https://orcid.org/0000-0001-7384-7972

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